Building Science

Lecture 12: Building energy fundamentals

Dr. Matthew Francis
School of Property, Construction & Project Management
RMIT University, Melbourne, Australia

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Lecture outline

- Introduction to electricity in Buildings
- Energy Sources for Buildings
- Solar & Wind power

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Energy

What are the 2 categories of Energy used in Buildings?

Where does energy for buildings come from?

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Energy

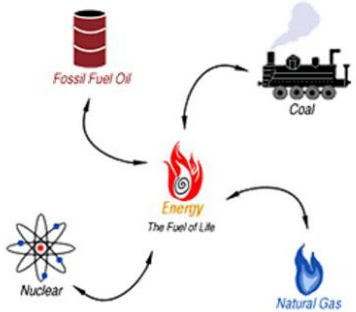
Energy Sources

RENEWABLE

- SOLAR
- WIND
- BIOMASS
- HYDRO POWER
- MARINE POWER
- GEOTHERMAL

Non-RENEWABLE

- FOSSIL FUELS
  - *OIL*
  - *NATURAL GAS*
  - *COAL*
- NUCLEAR



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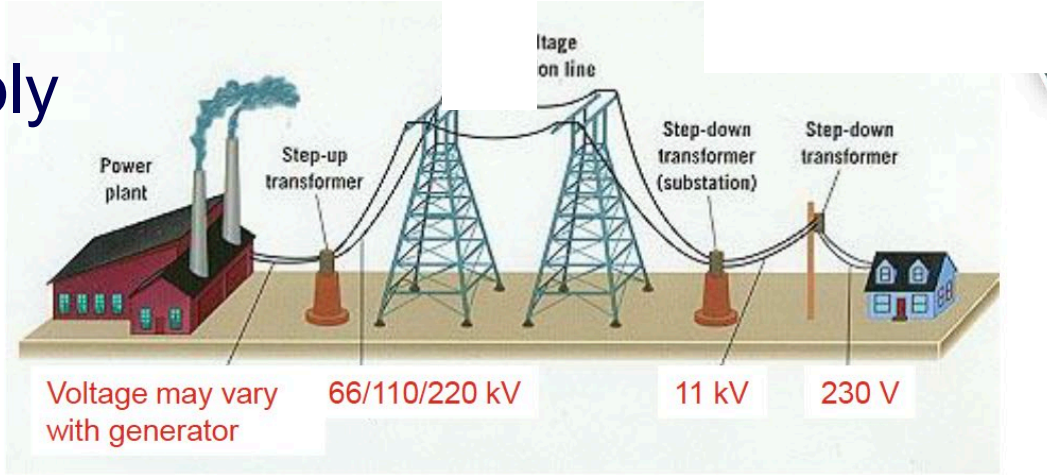
Electricity supply for buildings



Electricity transmission and distribution system is alternating voltage and alternating current (AC) – voltage varies between power plant and end load.

AC electricity is generated within a conductor when it's rotated within a magnetic fie when a magnetic field is rotated past a conductor.

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Energy

- **China's Pledge to the World:**
  - 碳达峰 (Peak Carbon): Reach peak CO_2 emissions before 2030.
  - 碳中和 (Carbon Neutrality): Achieve net-zero CO_2 emissions before 2060.

- **Why This Matters?**
  - The building & construction sector is a major contributor to energy use and emissions.
  - You are the key to achieving this goal. Your future designs directly impact China's national success.

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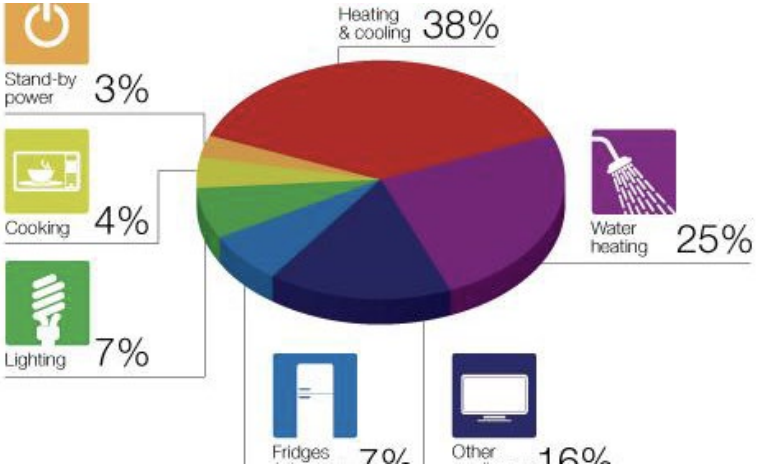
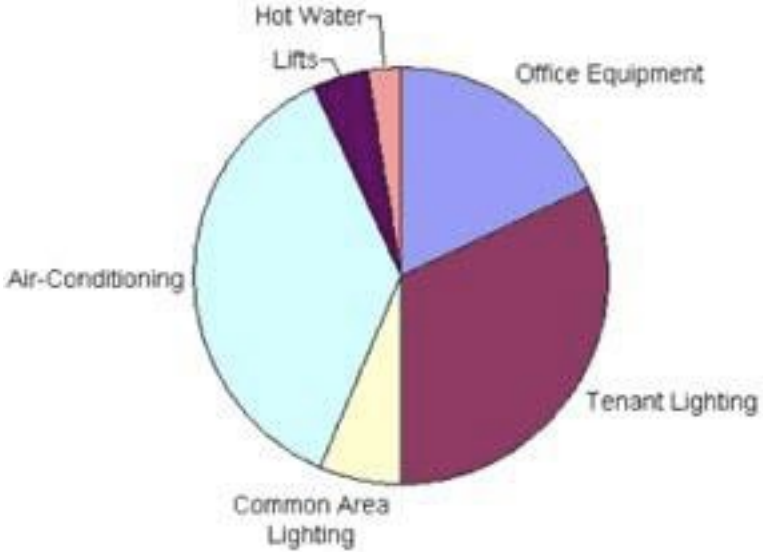
Energy

Typical home vs. Typical office?





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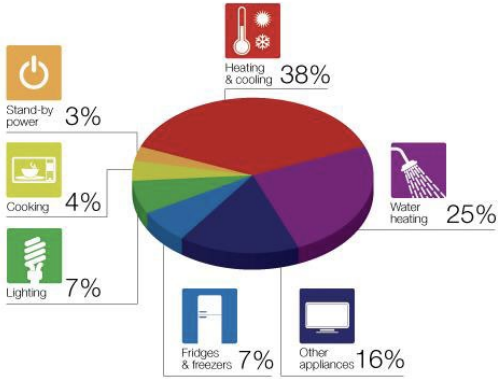
Energy

· **Core Principle:** Energy In = Energy Out ± Storage

· Our Job as Building construction managers: Manipulate this equation!

- Reduce unwanted "Energy Out" (heat loss/gain).
- Optimize "Energy In" (only use what we need).
- Utilize "Storage" (thermal mass) for stability.

· This is the foundation of all building physics.



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Energy – The golden rule

The Only Logical Approach to Design:

1. REDUCE LOADS (Passive Design principles)

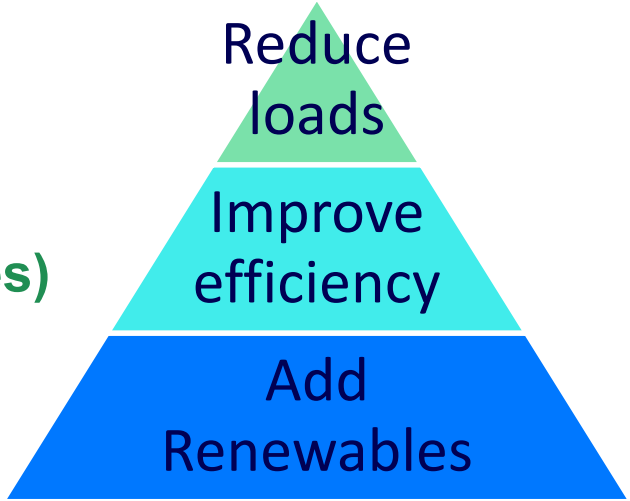
- Insulation, Airtightness, Shading, Orientation
- The most cost-effective step!

2. IMPROVE EFFICIENCY (Systems)

- High-Efficiency HVAC, Heat Pumps, LEDs

3. ADD RENEWABLES (On-Site Generation)

- Solar PV, Solar Thermal, Geothermal



> Always follow this sequence!

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Where is Energy consumed in buildings?

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Energy use in Buildings

- ✓ Energy is consumed by all buildings that:
  - ✓ To provide mechanical ventilation
  - ✓ HVAC and Air conditioning
  - ✓ Lighting and services
  - ✓ Lifts and escalators

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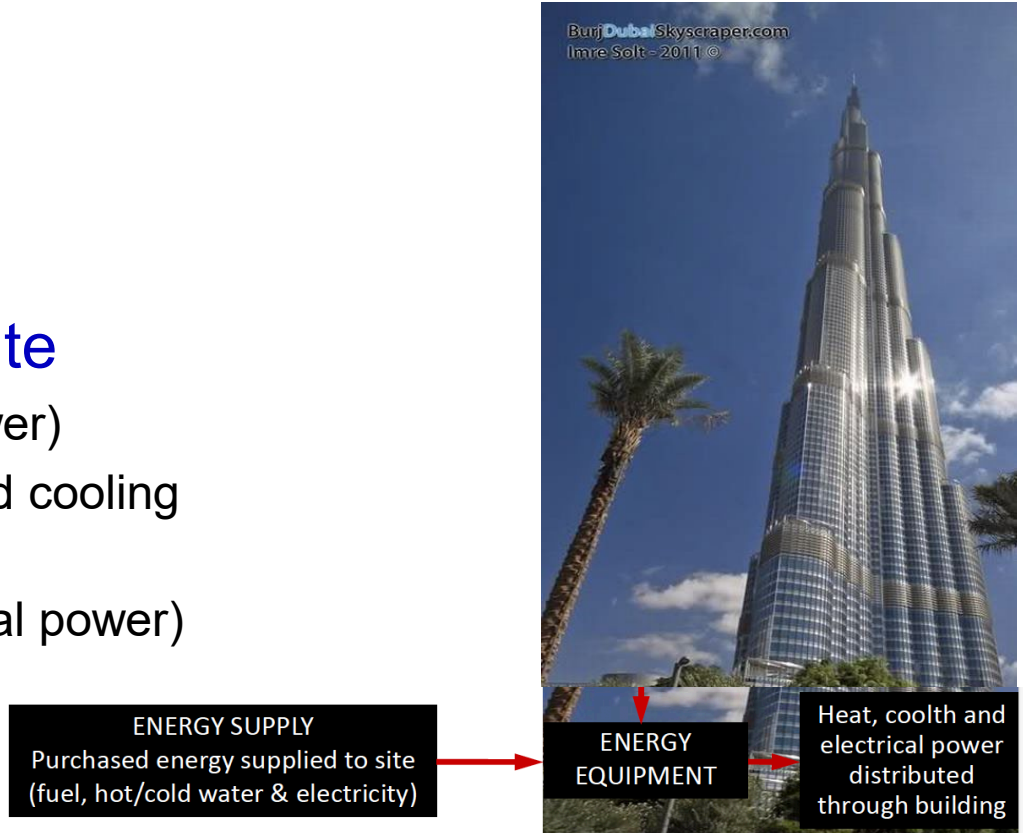
Energy supply systems

- ✓ Energy is supplied to site
  - ✓fuel (heat and electrical power)
  - ✓hot and cold water (heat and cooling respectively)
  - ✓electricity (heat and electrical power)



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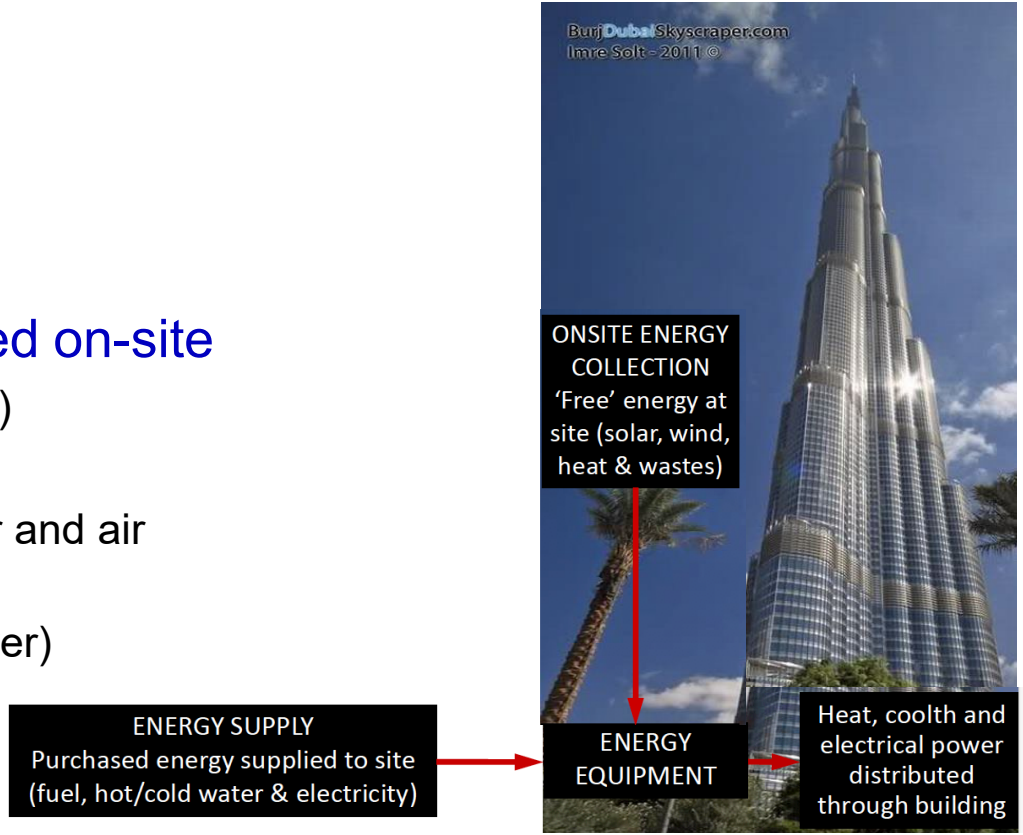
Energy supply systems

- ✓ Energy can also be produced on-site
  - ✓ solar (heat and electrical power)
  - ✓ wind (electrical power)
  - ✓ low grade heat in ground, water and air (heat)
  - ✓ wastes (heat and electrical power)



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Solar Energy

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Energy

Solar Energy

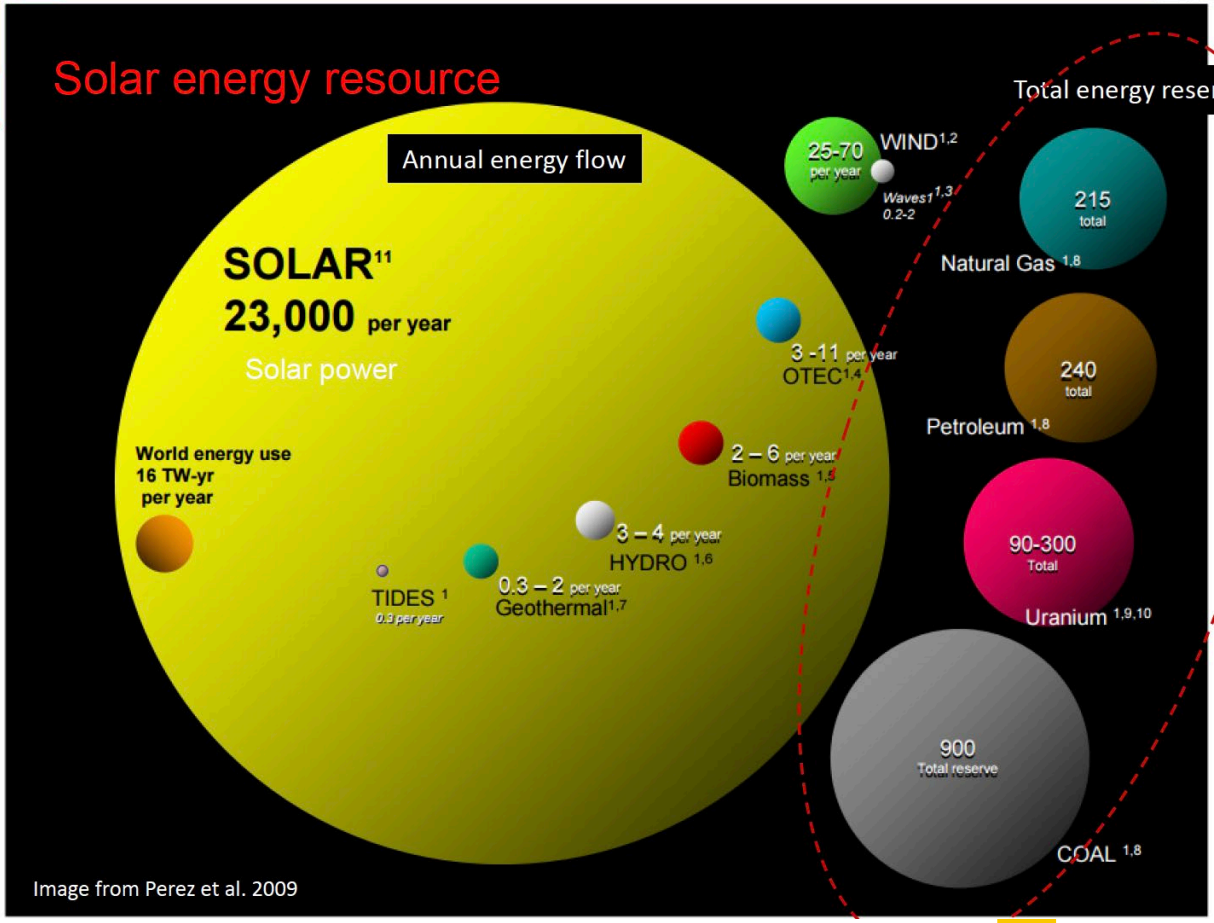
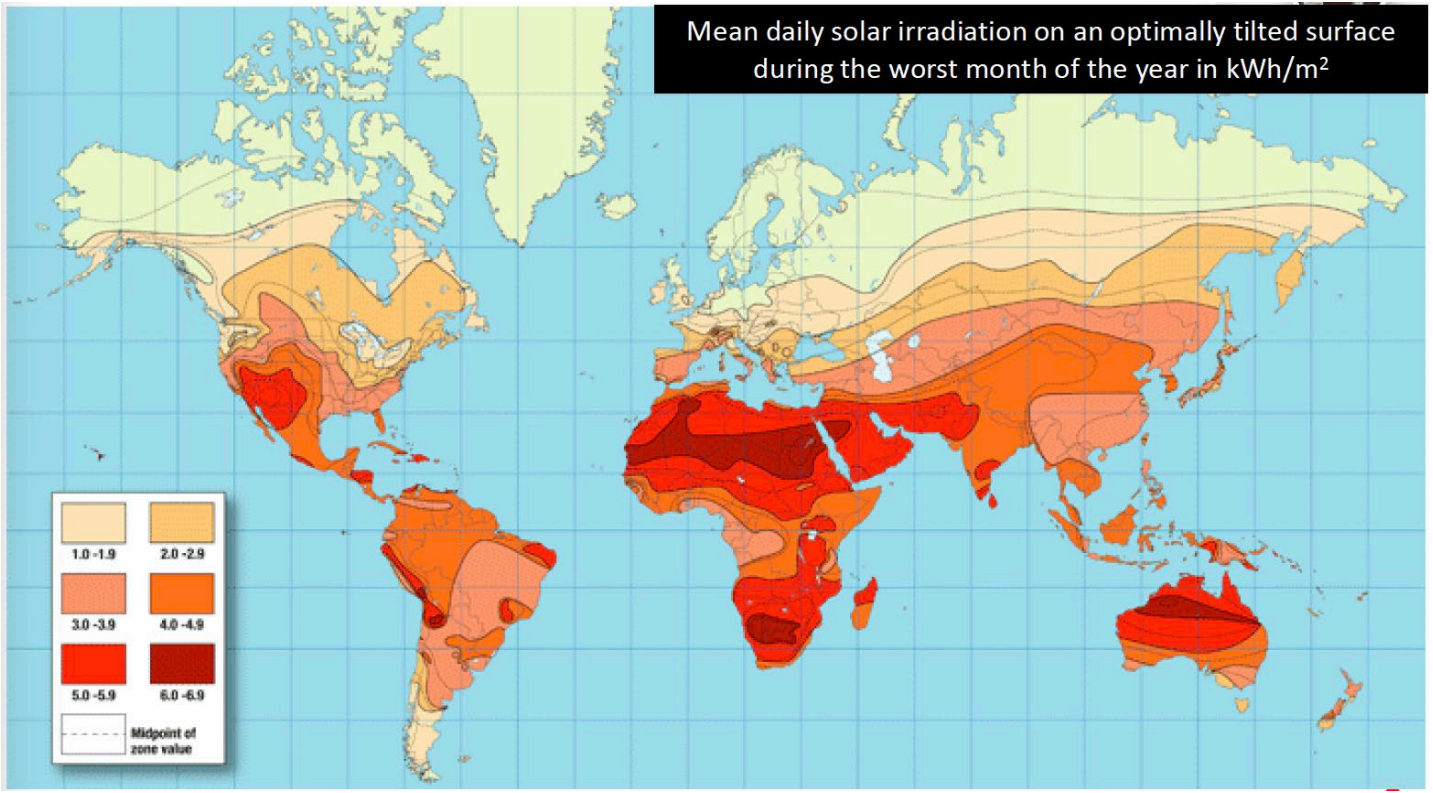


Image from Perez et al. 2009

Energy

Solar Energy



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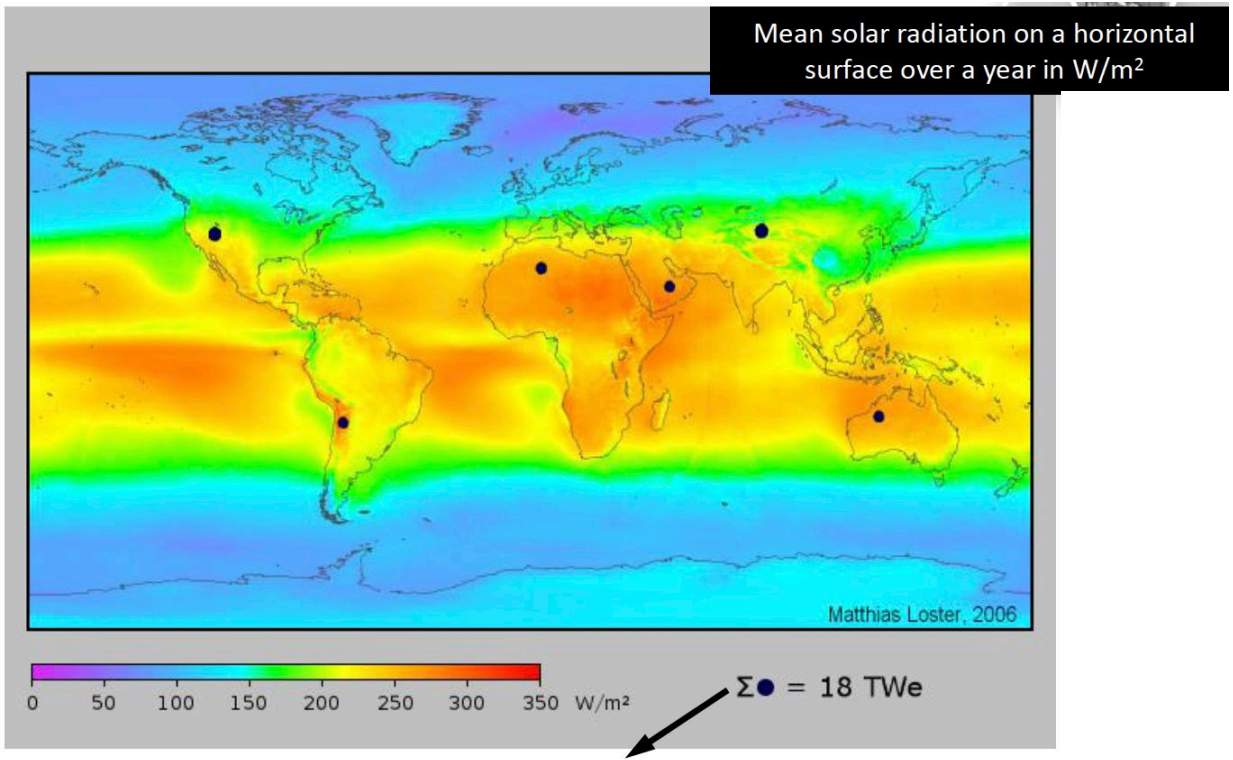
Solar Energy



Global energy demands could be supplied by ≅ 900,000 km² of photovoltaic panels with 8% efficiency

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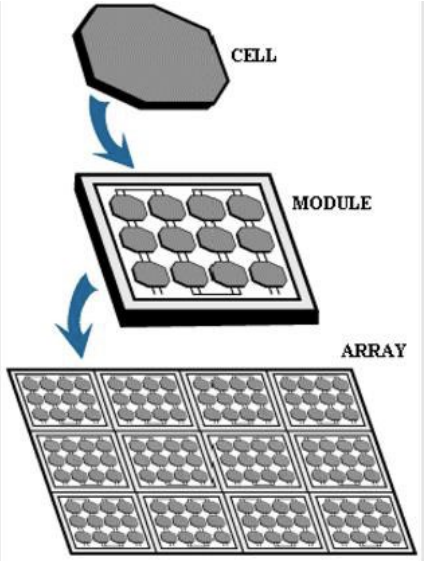
Energy

Solar Energy



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Solar Power
(Photovoltaic panels)

- ✓ How?
  - ✓ Convert solar energy directly to DC (direct current) electricity with efficiencies <15%
  - ✓ May be integrated into building

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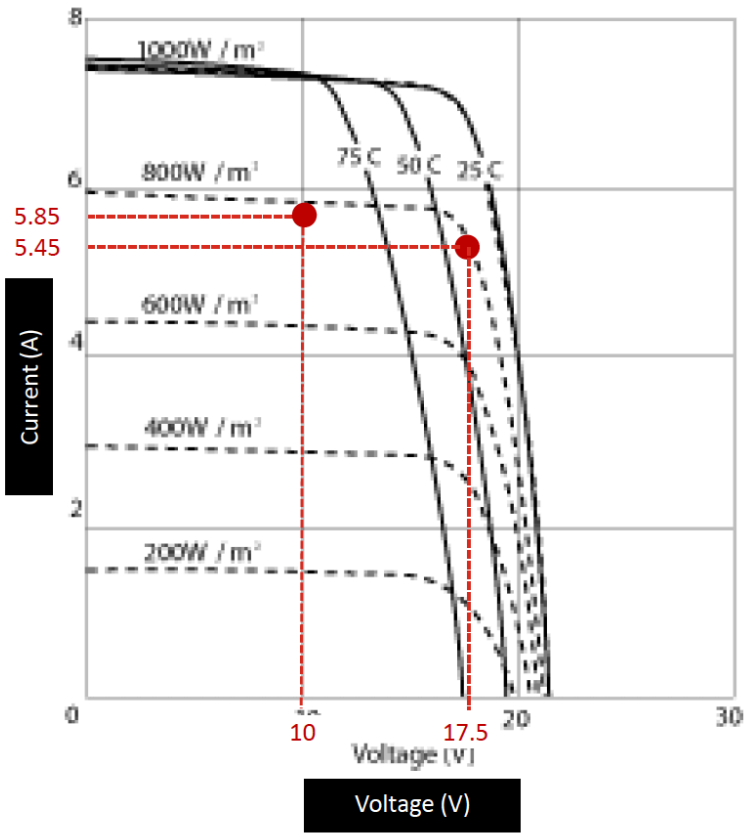
Example

Determine the efficiency of a PV panel with the performance characteristics shown if solar irradiation equals 800 W/m², panel temperature equals 25°C and:

(a) PV voltage equals 10 V
(b) PV voltage equals 17.5 V



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Solar Thermal Collectors

- ✓ Solar collectors use solar energy to heat air/water (e.g. absorption chiller)
- ✓ Efficiencies depend on ambient conditions, temperature of air/water and how well collector is insulated (0-80%)
- ✓ Collectors may be integrated into building or added later

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Example

Find the efficiency of a 50 m^2 solar water heater if solar irradiance equals 800 W/m^2, inlet and exit water temps equal 15°C and 45°C respectively and mass flow rate of water through the heater equals 0.2 kg/sec.

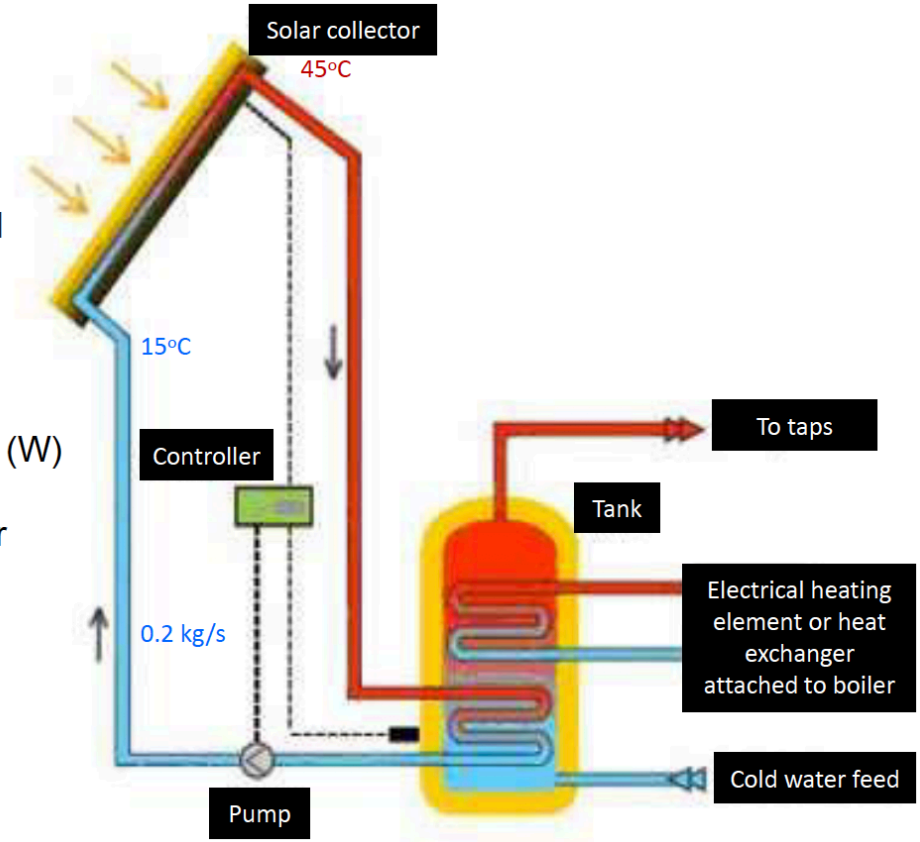
Thermal energy collected

$$= m \times C_p(T_{exit} - T_{inlet}) \quad (W)$$

Given the specific heat of water (C_p) equals 4190 J/(kgK) then:

$$\text{Efficiency} = \frac{0.2 \times 4190(45-15)}{50 \times 800}$$

$$= 0.63 \ (63\%)$$



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Wind Power

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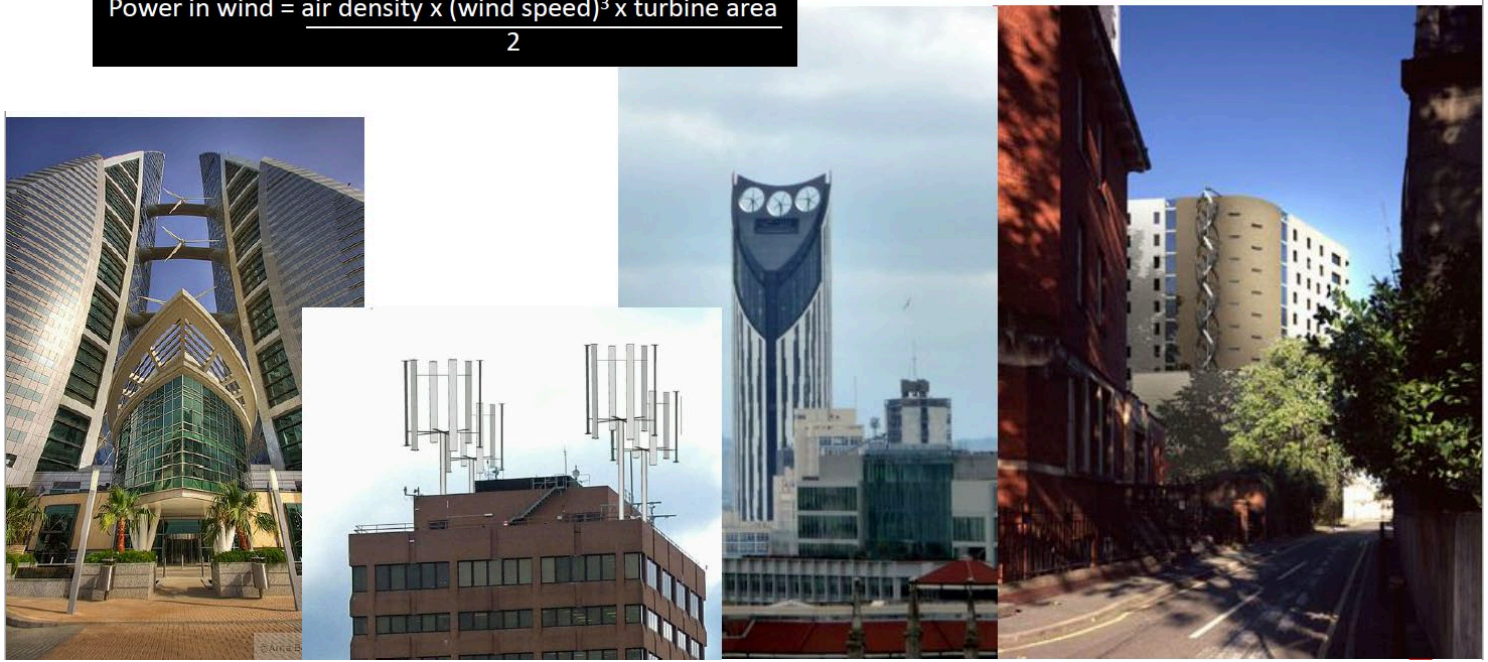
Wind Power

$$\text{Efficiency} = \frac{\text{Electrical power output}}{\text{Power in wind}}$$

$$\text{Power in wind} = \frac{\text{air density} \times (\text{wind speed})^3 \times \text{turbine area}}{2}$$



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Wind Power

Example

Determine the overall efficiency of a 5 m diameter horizontal axis wind turbine if the wind speed at the turbine equals 8 m/s, air density equals 1.2 kg/m^3 and electrical power output equals 2000 W.

Power in wind = $1.2 \times 8^3 \times (\pi \times 2.5^2)/2$
= 6,032 W

Efficiency = $\frac{2000}{6032}$ = 0.332 (33.2%)

Turbine efficiency typically less than 50%.

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Heat Pumps

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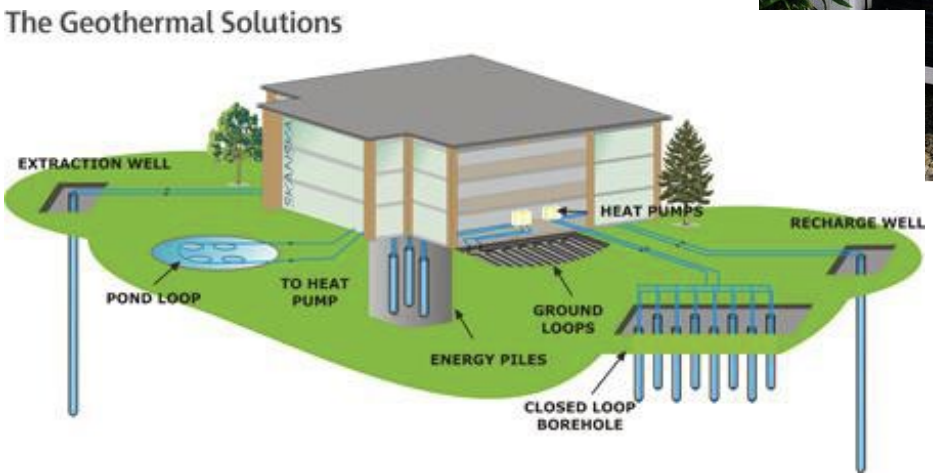
On-site Heat

- ✓ Direct heating/cooling
- ✓ Energy source/sink for heat pumps



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Energy in Buildings

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Energy – High performance envelopes

- The building envelope (façade) is our primary defence against the climate.

- **Five Key Properties of high performance buildings:**
  - Insulation (热绝缘): high R-Values – Resists conductive heat flow.
  - Airtightness (气密性): – Stops air leakage
  - Windows (窗户): low U-Values & good SHGC values (for optimal solar heat gain).
  - Adequate ventilation (with heat recovery) – 100% fresh air is ideal
  - Minimise all thermal bridges!

- **A good building envelope makes all other systems smaller and cheaper**

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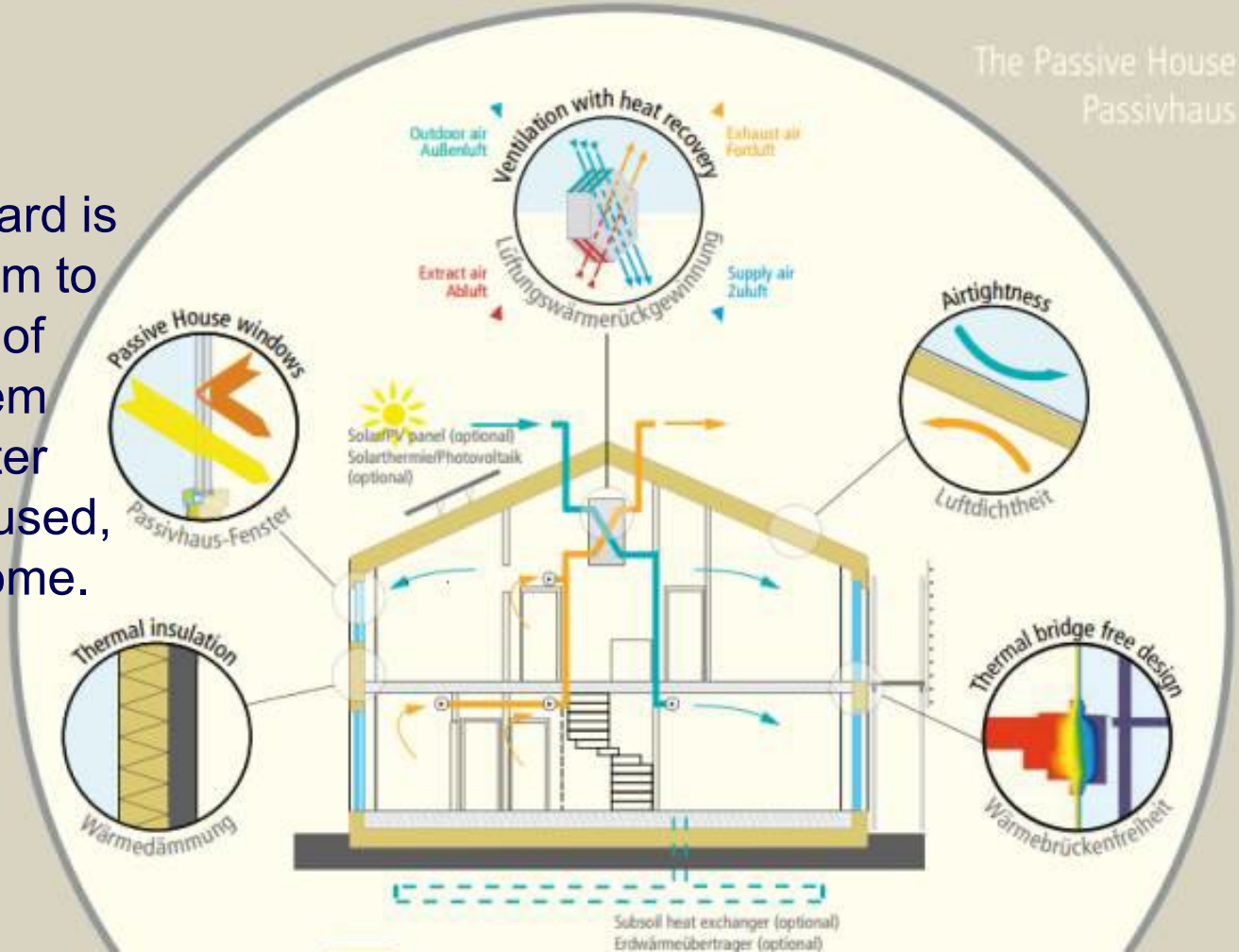
Passivhaus standard is an excellent system to use as it has all 5 of these in one system and does not matter what materials is used, just the final outcome.



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Energy – Ventilation & IAQ

- The Dilemma: An air-tight building saves energy... but can trap pollutants inside.

- The Solution: Mechanical Ventilation with Heat Recovery (HRV)
  - HRV (or ERV) (热回收通风系统): Devices that bring in fresh air and exhaust stale air while transferring heat between them.
  - Filter the air using HEPA filters.
  - Result: Excellent indoor air quality (IAQ) with minimal energy penalty.

- "Build Tight, Ventilate Right!" (建筑要密封，通风要高效!)

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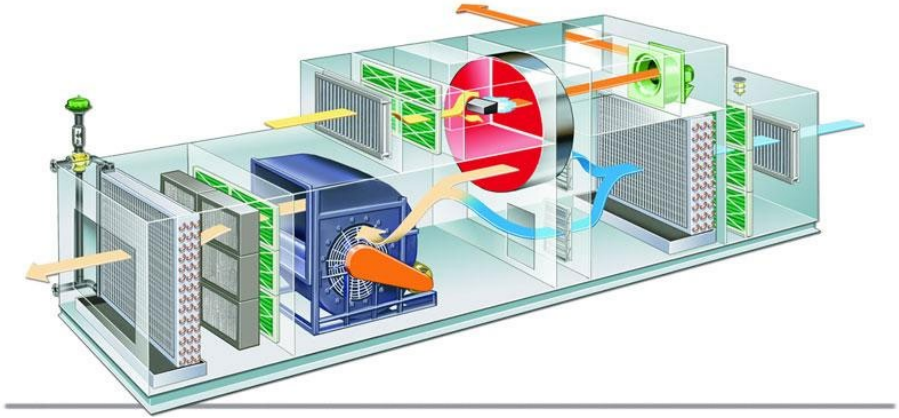
Energy – Ventilation & IAQ

Heat recovery ventilation (HRV)



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Energy – HVAC System efficiency

- How we measure system performance:
  - COP (性能系数): For heat pumps. A COP of 4.0 = 400% efficiency!
  - SEER (Seasonal Energy Efficiency Ratio) (季节能效比): For cooling efficiency across the entire year.
  - SEER = Cooling output (W.h) ÷ Electrical input (W.h)

- **Critical Rule:** Right-size the system!
  - An oversized HVAC system is inefficient, expensive, and uncomfortable.

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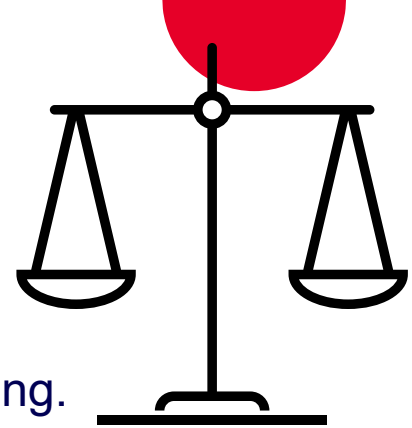
Energy – Carbon balance



- **Operational Carbon (运行碳排放):**
  - Emissions from heating, cooling, and powering the building.
- **Embodied Carbon (隐含碳排放):**
  - Emissions from manufacturing materials, construction, and demolition.
- **The Future:** As buildings become more efficient, Embodied Carbon becomes the dominant source of emissions.
- Therefore, material choices are crucial.

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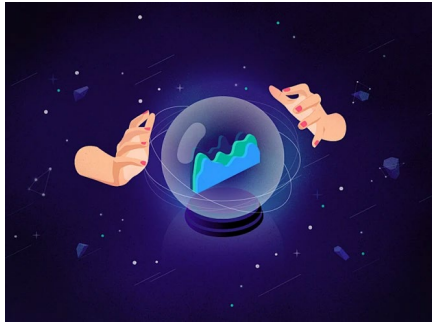
Energy – Building Energy Modelling

- What it is? Software that simulates a building's energy use before it's built.
  - Examples: ‘DesignBuilder, ‘PHPP’
- Why we use it?
  - Test design options and optimize performance.
  - Identify cost-effective strategies.
  - Comply with energy codes (e.g., GB standards).
  - Cheaper to change a model than the final building!
- **This is your most powerful design and analysis tool.**



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Energy – Case studies in China

Passivhaus in China:

- Northern China (Cold Climate):
  - Hebei Province (**Gaobeidian**): This is the epicenter of Chinese Passivhaus development. The city of Gaobeidian has become a living laboratory, hosting the "Railway Town" development, which includes the world's largest cluster of Passivhaus certified buildings (apartments, offices, a school, and a hotel).
  - This is a government-driven effort to demonstrate ultra-low energy building at scale.
- Eastern China (Temperate Climate):
  - Shanghai & Jiangsu Province: This economically advanced region has seen significant uptake. Examples include the **Qingpu Pinghe International School in Shanghai** (one of the largest Passivhaus schools in the world) and numerous projects in Nanjing and Changzhou.
  - The focus here is often on humidity control and cooling.
- Southern China (Hot Summer / Warm Winter Climate):
  - Guangdong Province & Fujian Province: Adoption is more recent but growing rapidly. The challenge here is fundamentally different: preventing overheating and managing extreme humidity.
  - Projects like the **One-Sea Public Welfare Academy in Shenzhen** demonstrate the standard's adaptability to hot climates.

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Energy – Case studies in China

- This is not just theory. It's happening here!
  - Shanghai Tower: Double-skin facade, high-efficiency HVAC & lift systems.
  - Passive House (Passivhaus) Standard: Ultra-low energy buildings in China's diverse climates.
  - Green Retrofits: Upgrading existing buildings for efficiency.
- **Your challenge:** To design the next generation of sustainable, resilient, and beautiful Chinese buildings.

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Thank you

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